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Linear eddy simulations of Reynolds number and Schmidt number effects on turbulent scalar mixing

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Effects of molecular diffusion on turbulent scalar mixing are studied using the linear eddy model. Unlike the energy spectrum which is determined only by the viscous and eddy time scales, the scalar spectra also depend on the diffusion time scales. The linear eddy model, being one-dimensional, offers an inexpensive way of capturing these time scale dependencies. Some of the observations made using experiments and direct simulations are verified using the model. Simulations reported here indicate that certain features of scalar mixing continue to depend on the Reynolds number (Re_λ) and the Schmidt number (Sc) in the range of parameter space ($32 \leqslant Re_\lambda \leqslant 775$ and $0.001 \leqslant Sc \leqslant 700$) that is unattainable using the current direct simulation capabilities. However, some of the models developed for differential diffusion using direct simulations are shown to be accurate even when the difference in the Schmidt numbers of the scalars and (or) the Reynolds numbers are very high. [DOI: 10.1063/1.1335540]

I. INTRODUCTION

Scalar mixing in turbulent flow is determined by two physical processes, turbulent transport and molecular diffusion. Turbulent transport is dominant at relatively large scales (i.e., production and inertial range scales) and owing to the effects of viscosity, become weak at the small scales (i.e., the dissipation scales). In contrast, molecular diffusion is predominant at the small scales. However, both processes interact at all length and time scales with no clear scale separation between them. Numerical investigation of these multiscale interactions is possible using direct numerical simulations (DNS). Although direct simulations have provided important physical insight[1–3] into scalar mixing, they are very expensive tools that are restricted only to low and moderate Reynolds numbers (Re). In the present study, an alternate and (relatively) inexpensive means based on the linear eddy model (LEM)[4] is used to explore issues concerning turbulent scalar mixing.

The inertial-convective range of scalar spectrum has a scaling law[5,6] that is similar to the inertial range scaling law for energy spectrum.[7] However, there is no similarity in the scalar and the energy spectra at smaller scales. The dissipation range is determined by viscous and eddy time scales and thus, viscosity uniquely determines its form in the high-Re limit.[8] In case of high wave number scalar spectrum, however, no universal form can be established due to the additional parameterization introduced by the diffusion time scale. The spectrum would depend both on viscosity and molecular diffusivity. There have been attempts to deduce scaling laws for high wave number spectra in very low[9] or very high[10–12] Schmidt number (Sc) limits, but these assertions are difficult to validate due to a lack of sufficient experimental evidence. The present study will attempt to address the effect of varying Sc on the scalar mixing process.

Scalar transport in turbulence simulations is often modeled using a gradient diffusion assumption. The total diffusivity in these models includes both molecular (D) and turbulent (D_t) diffusivities. Typically, D_t is related to ν_t (eddy viscosity) using a pre-specified turbulent Schmidt number Sc_t. At high turbulent intensities, $D_t \gg D$ and so the total diffusivities of both the scalars in a binary mixture would essentially be the same if Sc_t is the same. Under these conditions, differential diffusion effect cannot be taken into account in the model. However, molecular diffusion is known to play a significant role in transport of scalars even at very high Reynolds numbers.[13] When the molecular diffusivities of two chemical species are widely different (as for hydrogen and air), the evolution of their concentration fields may be different[13,14] at small scales. Recent DNS studies[1,3] have shed some light on this aspect of mixing for passive scalars. Studies such as these can help greatly in developing diffusion/mixing models for approaches like the probability density function (PDF) methods[15,16] and conditional moment closure (CMC) methods.[17] Some of the issues addressed in these DNS studies are revisited in the present work. Given that the one-dimensional linear eddy model is considerably cheaper than the DNS approach, Reynolds numbers and Schmidt numbers that are beyond the scope of DNS can and have been considered here.

The linear eddy model has already been used to predict both qualitative[18–20] and quantitative[21] features of turbulent scalar fields. This model is derived using the inertial range scaling law for the turbulent diffusivity. Though the scaling law is known, the exact value of the turbulent diffusivity (or a numerical constant that would determine its value) was necessary to obtain quantitative results in the past. In the present work, phenomenological arguments are provided to relate the constant that accompanies the scaling law to the Kolmogorov constant. Simulations are used to verify these arguments.

This paper is organized as follows. The model formula-

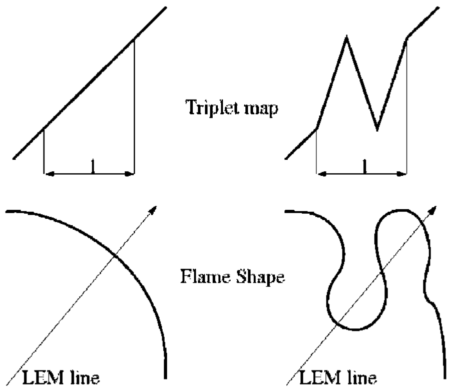

FIG. 1. Triplet map.

tion is discussed briefly in the next section followed by a description of the numerical implementation. In Sec. III, the results obtained over a range of Re and Sc are discussed with specific focus on the behavior of scalar spectra. This is followed by a discussion on the LEM and DNS predictions of differential diffusion among different scalars in turbulence. Finally, conclusions are presented in Sec. IV.

II. SIMULATION MODEL

The LEM provides a simple and inexpensive scheme to simulate the effect of isotropic turbulence on scalar fields. Though the present study is confined only to diffusive scalar fields (such as temperature and chemical concentrations), the model applies equally well to self-propagating scalar fields[21,22] such as those represented by the flamelet G equation.[23] The physical reasoning and the background behind the LEM have been reported earlier[24–26] and therefore, only the exact form of the model used in the current study is described here briefly.

A. Formulation

The LEM has two independent elements: molecular diffusion and turbulent stirring. Using Fick's law for molecular diffusion, the evolution for a scalar field, $\phi(x,t)$ can be represented by the following equation:

$$\frac{\partial \phi}{\partial t} = D\frac{\partial^2 \phi}{\partial x^2} + F_{\mathrm{stir}}. \tag{1}$$

In the above equation, F is a stochastic forcing term representing the effects of turbulent stirring. The effects of stirring are realized through an independent stochastic sequence of scalar field rearrangements. Of all the rearrangement schemes considered, a triplet mapping[25] is deemed to best represent the effect of inertial range turbulence on the scalar field. The schematic of the triplet mapping implemented on a scalar field with a uniform gradient is shown in Fig. 1. Each triplet map is assumed to represent the stirring of the scalar field by a single turbulent eddy. The geometric evolution of a scalar interface under the influence of a single eddy (the evolution triplet map models in one-dimension) is also illustrated in Fig. 1.

As seen in Fig. 1, the triplet map is fully conservative (conserves the mean value of the scalars) but increases the gradients threefold. The exponential growth rate of the scalar gradients by repeated (cumulative) multiplicative processes (triplet maps) is curtailed appropriately by the diffusion process. It is seen that the scalar gradient in isotropic turbulence is aligned most with the direction of most compressive strain rate.[27,28] The compressive effect of the turbulence on the scalar field, in the direction of its gradient, is emulated by the triplet map. So a possible interpretation[18] of the linear eddy domain is a space curve aligned with the local scalar gradient.

Both experiments[29] and simulations[28] have shown that, at least for $\mathrm{Sc}>1$ the scalar dissipation field is determined by diffusion counteracting discrete (spatially uniform and time varying) strain rate mechanisms that can be modeled in one dimension. In fact, a model similar to Eq. (1) for the scalar evolution along its normal direction was proposed by Yeung *et al.*[28] In this model, compression by a prescribed strain rate (with linear velocity variation along the flame normal) was used instead of the term F in the above equation. The strain rate field (F) in the current model is prescribed by a set of stochastic and discrete events determined by a Poisson process.

Since the turbulence is homogeneous, locations for implementing the triplet maps are chosen from a uniform distribution over the domain. The eddy size (l) for each triplet map lies between the Kolmogorov length scale, η, and the integral length scale, L, and is chosen from an eddy size distribution $f(l)$. The frequency of the triplet maps is determined by a Poisson process with an event parameter (stirring frequency per unit length of the spatial domain) λ which depends on the total intensity of turbulence. Finally, the size distribution, $f(l)$ depends on the distribution of energy between various scales of turbulence (i.e., the turbulent spectrum).

The material diffusivity $D_T(l_c)$ induced by a set triplet maps with length scales smaller than l_c is shown to be[4]

$$D_T(l_c) = \frac{2}{27}\lambda \int_{\eta}^{l_c} l^3 f(l)\, dl. \tag{2}$$

Note that D_T is a kinematic consequence of inertial range turbulence. Therefore, all scalars (i.e., temperature and chemical species) are affected in an identical manner.

Numerical experiments on flame broadening by Denet[30] are in agreement with the above arguments. In these experiments, the growths of temperature and specie concentration profiles due to stirring of a laminar flame by eddies smaller than the flame thickness were investigated. It was seen that, independent of the Lewis number (ratio of molecular diffusivities of thermal energy and mass), the thickness of the temperature and the species concentration profiles tend to become the same at high turbulent intensities. So the effect of fluid dynamic eddies on the dispersion of scalars is independent of their diffusivities. However, at scales much smaller than the (broadened) flame width, the effects of differential diffusion (caused by the difference in molecular diffusivities) are dominant and the spatial variations of the temperature and the species concentration need not be fully correlated.

Since $D_T(l_c)$ should be the same for all fluid properties (temperature, chemical concentrations, etc.), it should be independent of any properties specific to any one particular scalar in the flow. $D_T(l_c)$ has the same units as the $\nu_t(l_c)$, the eddy viscosity due to eddies smaller than l_c. Much like the ν_t, $D_T(l_c)$ (which results solely from stirring by fluid dynamic eddies in the inertial range) should only depend on the fluid dynamic characteristics of the inertial range (in addition to l_c). It is quite evident from the above (three) arguments that $D_T(l_c)$ is directly proportional to $\nu_t(l_c)$, the eddy viscosity due to the scales smaller than the length scale l_c. In traditional closure theories,[31,32] the eddy viscosity is used to close the generalized (kinetic) energy equation. Specifically, it used to close the term representing the turbulent diffusion of energy due to eddies smaller than a particular length scale, l_c. In that sense, ν_t corresponds to the energy diffusion rate (i.e., ν_t is D_T corresponding to a particular scalar, the kinetic energy). So as a first approximation, $D_T(l_c)$ is assumed to equal $\nu_t(l_c)$.

The mean (spatially averaged) eddy viscosity (ν_t) due to turbulence at scales smaller than l_c has been obtained using closure theories as follows:[31–33]

$$\nu_t(l_c)=0.441\alpha^{-3/2}\sqrt{\frac{E(k_c)}{k_c}}. \tag{3}$$

Here, k_c is the wave number corresponding to the length scale l_c and $E(k)$ is the energy spectrum. By assuming an inertial range form of the spectrum at the wave number k_c with α (Kolmogorov constant) of 1.5[34] and equating the eddy viscosity in Eq. (3)[31,35] to D_T, the following equation is obtained:

$$D_T(l_c)=0.064\nu\left[\frac{l_c}{\eta}\right]^{4/3}. \tag{4}$$

The above equation is the same as the inertial range scaling law that was used in earlier LEM mixing studies[4,24,25] except that the numerical constant (0.064) in the above expression is obtained here using the above arguments regarding the relationship between D_T and ν_t. Earlier LEM studies used the value of D_T obtained from either experiments[4] or DNS.[18–20] This was done either by evaluation of overall D_T (corresponding to the integral scale) or by matching the linear eddy time scale to the large eddy turnover time from DNS or experiments. With one exception,[4] these studies are not relevant to the current research since they are low-Re studies and a well defined inertial range may not exist. Hence, the LEM results of McMurtry *et al.*[18] did not fully match the corresponding DNS results. Realizing the fact that a proper inertial range does not exist in a low-Re DNS, Cremer *et al.*[19] designed a modified version of the LEM (called the narrow band formulation) that was found to better match those from a DNS. The current formulation, however, is designed for high-Re and estimates D_T based on turbulence phenomenology. The accuracy and the range of validity of this estimate is addressed in this paper by simulating high-Re flows and comparing with available experimental data.

All other features in the current LEM model are identical to the earlier[4] formulation and therefore, are summarized briefly. Thus, using Eqs. (4) and (2), and following the derivation procedure outlined by Kerstein,[4] equations for $f(l)$ and λ are obtained as

$$f(l)=\frac{5}{3}\,\frac{l^{-8/3}}{\eta^{-5/3}-L^{-5/3}}, \tag{5}$$

$$\lambda=0.6912\left[\frac{L}{\eta}\right]^{4/3}\frac{\nu}{L^3}\,\frac{[(L/\eta)^{5/3}-1]}{[1-(\eta/L)^{4/3}]}. \tag{6}$$

The numerical implementation of this model is described in the following section.

B. Numerical implementation

The one-dimensional diffusion equation (1) is solved numerically using a finite volume formulation. The diffusion time step is nominally $\Delta t_D=\Delta x^{-2}/D)$, while the time step (Δt_s) for the turbulent stirring is determined as the inverse of λL. Both processes occur at their respective time scales with turbulent stirring interrupting the diffusive process at the respective epoch time. Details of this process is given elsewhere[4,25] and therefore, avoided here for brevity.

The resolution requirement is determined as follows. The molecular Schmidt number (Sc) dictates the resolution required for accurate modeling of diffusion. Depending on Sc, the diffusion cutoff either equals $\epsilon^{-1/4}D^{3/4}$ (large Sc) or $\epsilon^{-1/4}D^{1/2}\nu^{1/4}$ (low Sc). The one-dimensional implementation of LEM enables adequate resolution of the diffusion cutoff with at least four grid points. However, there is another restriction on the size of the grid. The Kolmogorov length scale, η, is the smallest length scale at which the stirring is implemented using the triplet map (which requires six points). So the minimum resolution has to be no larger than one-sixth of η. Typically, a grid size of $\eta/12$ or $\eta/24$ is used in the simulations that are reported here.

When the scalar field is isotropic, periodic boundary conditions are used. In cases requiring statistical stationarity for time averaging over long times, stationary forcing is introduced at the low wave numbers. For cases where there is a mean scalar gradient, a linear profile is used as an initial condition for the scalar field and step periodic boundary conditions are imposed subsequently.

III. RESULTS AND DISCUSSION

The model presented in the previous section is used to predict the Obukhov–Corrsin constant that characterizes the inertial-convective range of scalar spectra. This prediction serves to provide a direct validation of the present formulation. The calibrated model is then used to study the effect of various nondimensional parameters such as Re and Sc on turbulent mixing.

The qualitative features of turbulent mixing are known to be well captured by the linear eddy model.[4,19,20,22,25,26] Since the focus of the current study is on the quantitative performance of LEM, only statistically stationary simulations are considered. Comparisons are made with both experiments and direct numerical simulations.

A. The Obukhov–Corrsin constant

Like the inertial range in energy spectrum, the inertial-convective range of the scalar spectrum is characterized by a $k^{-5/3}$ scaling law. Specifically, the scalar spectrum, $F(k)$, in this range has the following form:[36]

$$F(k)=\beta\epsilon^{-1/3}\chi k^{-5/3}. \tag{7}$$

The constant β in the above expression is called the Obukhov–Corrsin constant and $\chi(=2D\langle[\partial\phi/\partial x_k]^2\rangle)$ is the scalar dissipation. In recent times,[37] β_1 which is the constant of proportionality in the scaling law for one-dimensional scalar spectrum is referred to as the Obukhov–Corrsin constant. Obviously, β_1 is easier to compute in both experiments and simulations and is related to β by the relation $\beta_1=3\beta/5$.

The scalar dissipation rate χ and β, together determine the intensity of the spectrum. The accurate prediction of both these quantities, simultaneously, can serve as a quantitative measure of the accuracy of the present LEM simulation model. An isotropic, statistically stationary scalar field is used here. The turbulent Reynolds number is set to 20 500 ($\mathrm{Re}_\lambda\sim550$), which corresponds to about 1680 Kolmogorov (η) length scales per each integral length. These conditions are at a sufficiently high Re to establish an unambiguous inertial range.

Low wave number isotropic forcing is used to achieve statistical stationarity of the scalar field. Statistical stationarity of the scalar field can also be achieved by using a mean scalar gradient in stationary isotropic turbulence or near stationary isotropic turbulence (as in weakly decaying grid turbulence). This method of achieving isotropy is not used here since there is increasing evidence[38–40] that in the presence of large scale (or the mean) scalar gradients, the scalar field has some anisotropy even at the smallest of scales. The ideas of local small scale isotropy and Obukhov–Corrsin scaling (and the inertial scaling for the energy spectrum) have historically been viewed as direct consequences of the cascade picture of turbulent scales. It is not clear what role the deviation from isotropy (i.e., anisotropy) plays in determining the scaling law (or the correction to the Obukhov–Corrsin spectrum). In light of these uncertainties, a scalar field that is driven by large scaling isotropic forcing (and as a result, has isotropic large scales) is deemed a better choice for determining β.

After a stationary state is achieved, the value of the scalar dissipation is computed using the scalar field. In LEM, χ equals $2D\langle[\partial\phi/\partial s\rangle]^2$ since the coordinate ''s'' in the linear eddy domain is aligned with the scalar gradient. The scalar spectrum [Eq. (7)] computed from the stationary state is shown in Fig. 2, in a normalized form. The normalization is done as required by the inertial convective scaling law in order to produce a plateau at the inertial convective wave numbers. The magnitude of the plateau is the scaling constant, β.

Although the (scaled) spectrum shown in Fig. 2 is computed using one-dimensional linear eddy model results, the

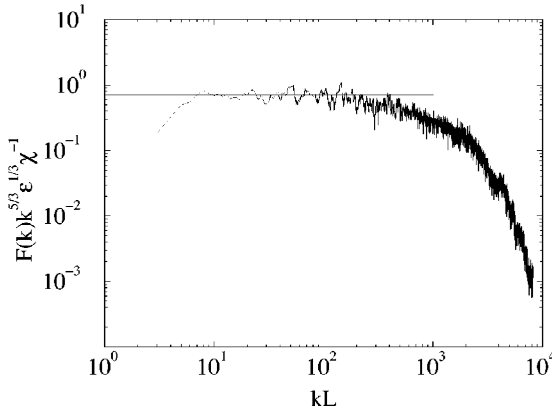

FIG. 2. Normalized scalar spectrum at $\mathrm{Re}_L\sim20\,500$. The plateau that is created due to this nondimensionalization indicates the value of the Obukhov–Corrsin constant.

plateau corresponds to the three-dimensional Obukhov–Corrsin constant β instead of the 1D counterpart β_1. This is easily explained as follows. First, ν_t corresponding to three-dimensional turbulence (which differs from that of two-dimensional turbulence[31] or one-dimensional turbulence) is used to arrive at the model used here. The predicted results therefore reflect the three-dimensional nature of turbulent mixing. This fact can easily be established using the fact that the linear eddy domain is aligned with the scalar gradient and all the scalar dissipation occurs on this domain.

Consider the following exact expressions for the scalar dissipation spectrum and the total scalar dissipation in three-dimensions:

$$\chi'(k)=2Dk^2F(k), \tag{8}$$

$$\chi=\int\chi'(k)dk. \tag{9}$$

Here, k is the scalar wave number, i.e., $k=(k_1^2+k_2^2+k_3^2)^{1/2}$. Now consider the equivalent expressions in the linear eddy formulation:

$$\chi'_{\mathrm{LEM}}(k_{\mathrm{LEM}})=2Dk^2F_{\mathrm{LEM}}(k_{\mathrm{LEM}}), \tag{10}$$

$$\chi_{\mathrm{LEM}}=\int\chi'_{\mathrm{LEM}}(k_{\mathrm{LEM}})dk_{\mathrm{LEM}}. \tag{11}$$

Here, k_{LEM} is the wave number corresponding to the one-dimensional spatial coordinate ''s'' in the model. The difference between the scalar dissipation, χ_{LEM} computed using Eq. (11) and the exact scalar dissipation, χ [computed using experimental data[39] and Eq. (9)] is found to be less than 4%. Also, the integrand $\chi'(k)$ in Eq. (8) and $\chi'_{\mathrm{LEM}}(k_{\mathrm{LEM}})$ in Eq. (10) have the same $k^{-5/3}$ scaling. As a result, it can be concluded that the model scalar dissipation spectrum in Eq. (10) corresponds to the scalar wave number dependent scalar dissipation spectrum in Eq. (8). By comparing Eqs. (10) and (8), it is concluded that the spectrum $F_{\mathrm{LEM}}(k_{\mathrm{LEM}})$, which is shown in Fig. 2 in a scaled form, corresponds to the three-dimensional scalar spectrum written as a function of scalar wave number [i.e., $F(k)$] as in Eq. (7). Hence, the plateau in Fig. 2 corresponds to β and not β_1. As shown in Fig. 2 by a horizontal line, the predicted value of β

equals 0.7. Also, the subscript ''LEM'' is dropped in further presentations noting the fact the model spectrum corresponds to a three-dimensional scalar spectrum.

For comparison with values from earlier DNS and experimental studies, the value of β_1 is evaluated and it is found to be 0.42. A discussion of the inertial convective range of scalar spectra using an extensive compilation of scalar mixing data was presented by Sreenivasan.[37] This review concludes that the value of an universal Obukhov–Corrsin constant (if one exists), is around 0.4. A more recent experimental study using grid turbulence in the presence of a mean scalar gradient by Mydlarski and Warhaft[39] quotes a range of 0.45 to 0.55 for β_1. Some of the scalar spectra measured by Mydlarski and Warhaft[39] do not show a clear $-5/3$ spectral scaling at all inertial-convective wave numbers. The high wave number end of the inertial range is characterized by a logarithmic spectral slope larger than $-5/3$ (i.e., spectrum scaled as in Fig. 2 shows a bump at the high wave number end of the plateau). There seems to be no explanation for this bump (termed as the bottleneck phenomenon by Warhaft[40]) as yet, but their presence does seem to interfere with the determination of the Obukhov–Corrsin constant. However, there is still a lot of scatter in the values of β_1 that are computed from spectra, which have a clear inertial-convective scaling.

A possible explanation is as follows. It is very likely that the uncertainty in the actual value for the Obukhov–Corrsin constant might be related to the discrepancies associated with determination of the Kolmogorov constant. Sreenivasan also concluded earlier[34] that the most likely value of α is 1.5, which gives a scaling constant (α_1) for one-dimensional energy spectrum a value of 0.49. The constants α_1 (specified) and β_1 (obtained using LEM) in this study are close to their corresponding values given by Sreenivasan. The scatter in $\alpha_1(=18/55\alpha)$ from 0.45 to 0.66[34] could easily explain the scatter of β_1 from 0.36 to 0.533, if a linear relationship[37] (i.e., $\beta_1 \sim 0.8\alpha$), as predicted by LEM, is assumed. The assumption made in this argument is that the scaling laws for energy and scalar spectra result, at least in part, from a similar source (i.e., the turbulent stirring).

Although the momentum field and the scalar field are advected (stirred by turbulence) in a similar fashion, they evolve differently due to the differences between the viscous and diffusive effects. In addition, the solenoidal constraint (divergence-free condition) might also affect the evolution of the momentum field. As a result, velocity and scalar fields might, sometimes, have distinct characteristics (such as intermittency and spectral scaling). But, the fact remains that the intensity of the energy spectrum and the scalar spectrum (which respectively determine the total mechanical energy and the scalar energy) depend on the magnitude of stirring. Since the magnitude of stirring depends on α (which determines the extent of energy in the inertial range eddies), it is reasonable to expect that β_1 and α_1 are related in a monotonic way. This argument, though plausible, might seem to be an oversimplified and a very deterministic view of turbulent mixing. It would, therefore, be enlightening to see if the experiments with higher and lower values of β_1 correspond, respectively, to higher and lower values of α_1.

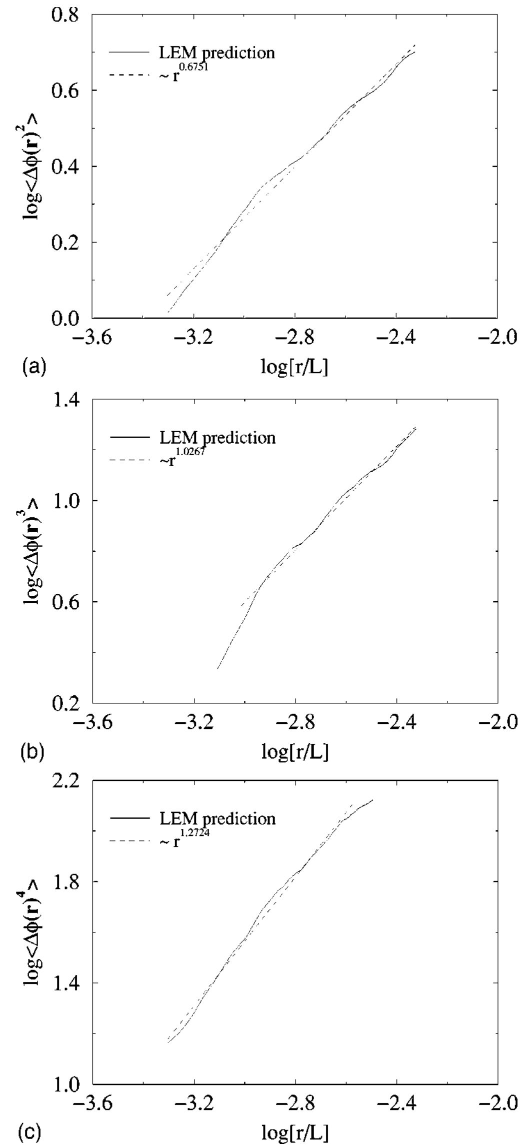

FIG. 3. Structure functions of the scalar field. Kolmogorov hypothesis: nth order structure function $\sim r^{n/3}$. The curve fits to LEM predictions show exponents very close to these exponents.

Presented in Fig. 3 are (a) second, (b) third and (c) fourth order scalar structure functions produced by LEM. A nth order structure function is defined as $\langle |u(x)-u(x+r)|^n \rangle$ with argument r, the separation distance. The curve fits to the predictions in the form of r^a type functions are also shown. The second and third order structure functions are predicted as per the Kolmogorov hypothesis (extended to scalars). The fourth order structure function is less close to the required behavior. This deviation represents the presence of intermittency in the solution, despite the fact that intermittency corrections to the inertial range scaling laws (for the velocity or the energy spectra) have not been included in the design of the linear eddy model. Holzer and Siggia[38] have also reported that scalar fields can develop intermittency even in purely Gaussian velocity fields. The current prediction (in a statistical sense) is in agreement with their observation. However, as pointed out by Yeung,[3] it is unlikely that the intermittency in the linear eddy model would be similar to that of the Navier–Stokes turbulence. This, in part, could be due to the fact that all scalar dissipation occurs on a one-dimensional domain in the linear eddy model (when compared to the three-dimensional space in Navier–Stokes turbulence).

Besides the physical features like intermittency, there are some numerical and implementational aspects of modeling that can lead to variations in higher order statistics. Wunsch[41] used a LEM type model and found that the predictions of exponents of higher order correlations (structure functions) are affected by the finite computational domain size. The reason for this is that although the domain size is larger than the integral length scale, the largest eddy size is of the order of the integral length scale. Thus, the length scale PDF, $f(l)$, is discontinuous at the integral length scale, L, and as a result, although turbulent stirring is probable at length scale, L, it is not at a slightly higher length scale. This was shown[41] to lead to the formation of exponential tails in the scalar difference probability density functions which affect the higher order statistics more than the lower order ones. It has been argued that the domain size be at least 8 times the integral length scale and Re_λ be at least 30 before the tails start to appear. Both conditions are met in the simulations here and hence the higher order statistics could be affected by these exponential tails.

Hou *et al.*[42] found that the scaling exponents in three dimensions are also dependent on the domain size and as a result, power laws in physical space (e.g., Kolmogorov scaling for structure functions) are not easily recoverable from the power laws in spectral space [e.g., $E(k)$, $F(k)\ k^{-5/3}$]. So, the effects of finite domain sizes discussed above are not unique to LEM type models. Hou *et al.*[42] also noted that a power law that is valid over a finite range of scales, say η to L, does not necessarily imply the validity of the equivalent spectral scaling over the corresponding wave number range (i.e., k_η to k_L, and vice versa). Such discrepancies between spectrum observations and second order structure function scaling have also been noted and discussed by Warhaft[40] who suggests a possible link between this discrepancy and the spectral bumps (bottleneck phenomena) discussed earlier in the context of determining value of β_1. Hou *et al.*[42] argue that for homogeneous turbulence, the spectral representation is more intrinsic than the spatial representation (i.e., validity of the spectral scaling law does not immediately imply that the structure functions of all orders in physical space scale as expected). Since the simulation here corresponds to homogeneous turbulence in a periodic box, one can expect[42] a wider range of applicability of (Kolmogorov type) Obukhov–Corrsin scaling in spectral space (as in Fig. 2) than in physical space (see Fig. 3). Given the observations of Wunsch[41] and Hou *et al.*,[42] caution is necessary while deriving conclusions (about features like intermittency) from the scaling exponents of higher-order structure functions.

B. Re and Sc dependencies

An analysis of Re effects on turbulent mixing is attempted here by simulation of experiments conducted by Mydlarski and Warhaft.[39] In these experiments, grid turbu-

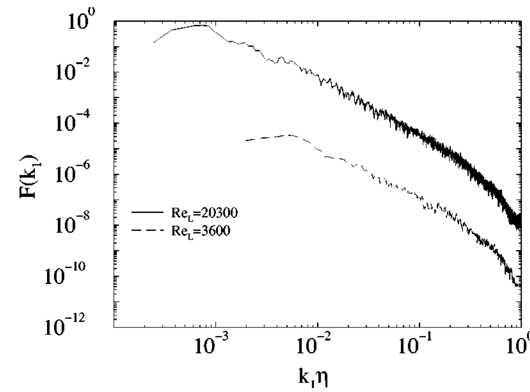

FIG. 4. Scalar spectra at moderate and high Re_L. Significant inertial-convective range is seen at both Re.

lence is used to transport a scalar field with a mean gradient in one of the transverse directions. As stated earlier, step periodic boundary conditions are used in the LEM in order to provide the source for scalar fluctuations in the simulations. Since the LEM domain aligns with the scalar gradient locally, the overall direction (i.e., the line between the end points) is assumed to align with the direction of mean gradient. The decay rate in the grid turbulence is assumed to be small and the turbulence is assumed to be nearly stationary in order to determine stirring parameters in LEM. The parameters characterizing this stationary turbulence (L, u', η) are set to their corresponding values in experiment, measured at the same downstream location (from the grid) where the scalar measurements are taken. The scalar spectra at Reynolds numbers (Re_L) of 3600 and 20 300 are shown in Fig. 4. The graph corresponding to Re_L of 3600 is shifted two decades down on the ordinate-axis for clarity. Both spectra have significant inertial-convective ranges.

The logarithmic spectral slopes in the inertial range are predicted with less than 4% deviation from required scaling for Re_λ larger than 300.0 ($\mathrm{Re}_l \sim 5600$). In case of experiments, a $k^{-5/3}$ scaling for the scalar spectra is not well established for $\mathrm{Re}_L \leqslant 5600$. This discrepancy between the measured and the predicted scaling exponents can be explained as follows.

Let n_1 and n_θ, respectively, denote the exponents in the scaling laws for the inertial range energy spectrum and the inertial-convective range scalar spectrum (i.e., energy spectrum $\sim k^{-n_1}$ and scalar spectrum $\sim k^{n_\theta}$). At a fixed Sc, both n_1 and n_θ tend to 5/3 monotonically with increasing Re. At lower (moderate) Re, both have lesser values and either one of them could be closer to 5/3 than the other. Most experiments[37,39] seem to suggest that n_θ is often closer to 5/3 than n_1 at moderate Re. In the experiments of Mydlarski and Warhaft,[39] a clear $k^{-5/3}$ scaling was never seen even for high Re. Despite this fact, a model based on a $k^{-5/3}$ scaling for the inertial range spectrum is used here. The use of this scaling (even at low Re) has led to logarithmic scalar spectral slopes that are much closer to 1.667 than in case of experiments.[39]

At first glance, the model seems to predict results that are somewhat contradictory to the fact that n_θ is larger (and closer to 5/3) than n_1. However, it is to be noted n_1 is

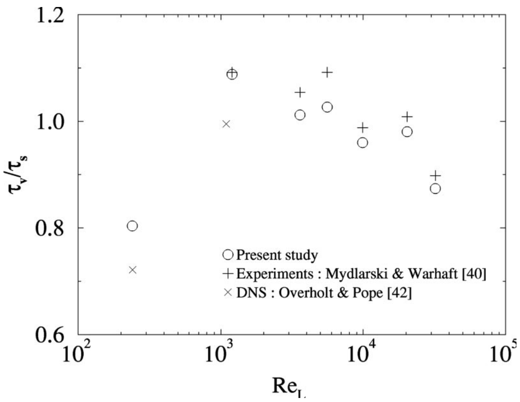

FIG. 5. Ratio of time scales characterizing kinetic energy and scalar decay. Since all other quantities used in the definition of time scale ratio are inputs, the graphs actually shows the prediction of scalar dissipation by LEM.

prescribed in the model and n_θ is predicted. So, LEM does not automatically ensure that n_θ is larger or any closer to 5/3 than the prescribed value of n_1. It is possible to construct a linear eddy models for arbitrary values of n_1 but it is difficult to calibrate them. The Kolmogorov constant accompanies a $k^{-5/3}$ scaling law and has an established range of values. There is no established way of providing numerical constants of proportionality for more general k^{-n_1} type scaling laws. Without the constant of proportionality, it is not possible to determine λ, the stirring parameter (rate).

The calculated and measured values of ratio of time scales characterizing energy decay (τ_v) and scalar decay (τ_s) are shown in Fig. 5. The mathematical expression for this ratio is given as

$$\frac{\tau_v}{\tau_s}=\frac{u'^2}{\epsilon}\frac{\chi}{\phi'^2}. \tag{12}$$

In the above equation, ϕ'^2 is the variance of the scalar ϕ. Note the use of u'^2 in the definition of the time scale ratio here instead of the more often used quantity q'^2. The turbulence information is part of the input, so the ratio of time scales actually measures the scalar dissipation rates. Also shown in Fig. 5 are the DNS predictions of Overholt and Pope[2] at moderate Reynolds numbers ($\mathrm{Re}_L \sim 1000$). The Re_L in the equation for λ [Eq. (6)] is defined using a turbulent length scale, i.e., $L \sim u'^3/\epsilon$ which in turn leads to the $\mathrm{Re}_L = [L/\eta]^{4/3}$ scaling. On the other hand, the integral length scale used to define the Reynolds number in the DNS study[2] corresponds to an average correlation length scale and is quite different from the turbulent length scale. So the Reynolds number, Re_L for the DNS simulations has been redefined with a length scale that equals u'^3/ϵ.

Stationary turbulence is used to stir the scalar field in this DNS. So it is quite similar to conditions in LEM. While the LEM predictions compare very favorably (to within 6%–8%) with experiments, there is some overprediction of χ compared to corresponding DNS results.

It needs to be remembered that the LEM stirring is confined to the scales between the integral length scale and the Kolmogorov length scale. The energy spectrum is assumed to be of the inertial form in all of these scales. On the other hand, stirring by scales larger than the integral length scale is possible in a DNS. The energy spectrum in the direct simulations[2] also differed significantly from the experiments of Comte-Bellot and Corrsin[43] at low wave numbers. As pointed out by Overholt and Pope,[2] the stochastic forcing used to achieve stationarity of turbulence could have an effect on scalar fluctuations. The scalar dissipation, on the other hand, is likely to be dominated by the small scale dynamics and may not depend strongly on the low wave numbers.

The increase of the time scale ratio with Reynolds number as seen in DNS, is not indefinite. In the range of Re_L where DNS is not possible, the time scale ratio does saturate with Reynolds number and actually shows a decreasing trend,[39] which is captured by the model. The predictions also seem to echo the experimental nonmonotonic relationship between the time scale ratio and the Reynolds number. This behavior could have resulted from the fact that the Reynolds number is varied by varying both the integral length scale and the turbulent intensity in a random (nonmonotonic) fashion. The experiments indicate that the time scale ratio is not solely a function of the flow Re but might depend on some other parameter that is not fully evident at present. One likely parameter is the ratio of the domain size to the integral length scale of turbulence. This ratio would determine the contribution of scales larger than the integral length scale to the fluctuation intensities. As mentioned earlier, this ratio can have some impact on the scalar fields if it is less than 8.[40] The decay rate is known to depend on the ratio of velocity to scalar length scale.[40] But in homogeneous flows with scalar gradients, it is difficult to define a length scale for the scalar field. However, this is true only if the scalar field is infinitely large. In reality (confined experiments and simulations), the domain size does restrict the size of the largest structure in the scalar field. If the mean scalar field (with uniform gradient) is subtracted out of the scalar field in the experiments, a length scale can be defined for the remaining field and is likely to be a function of the integral length scale of turbulence and the domain size. In this sense, the ratio of domain size to the integral length scale could also be related to the ratio between the velocity and scalar length scales.

The scalar fluctuation levels at length scales larger than the integral length scale (or equivalently, the spectrum at wave numbers lower than the integral wave number) in the model is determined by backscatter. The model simulations use the same set of parameters as in the experiments and the good match in the time scale ratios is encouraging despite the lack of a complete explanation or an understanding of backscatter in the model.

The results that have been discussed so far, are for the temperature of air so the Prandtl number (Sc equivalent for temperature) is 0.7. While the inertial-convective scaling is insensitive to Sc, the spectrum at wave numbers higher than those in the inertial range is highly Sc dependent. The forms of scalar spectra at very high Sc and very low Sc were hypothesized by Batchelor[10] and Batchelor, Howells, and Townsend,[9] respectively.

A stationary turbulent flow with Re_L of 10^4 and Sc of 10^3 is simulated using LEM. As expected, an inertial-

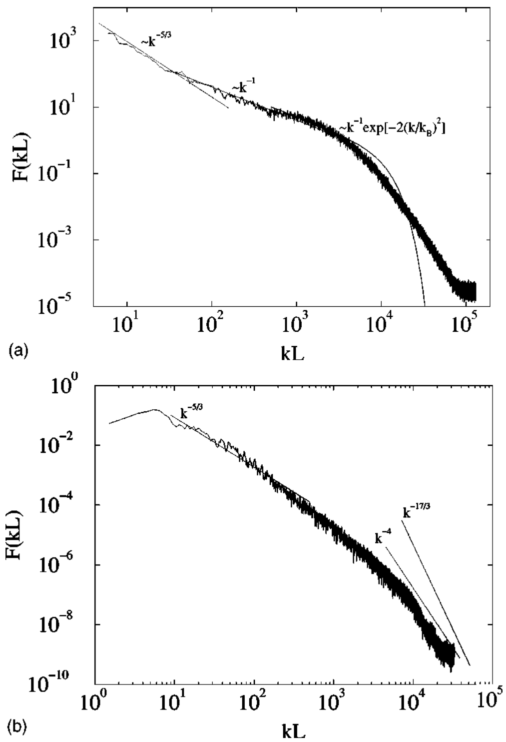

FIG. 6. Sc dependency of high wave number spectra.

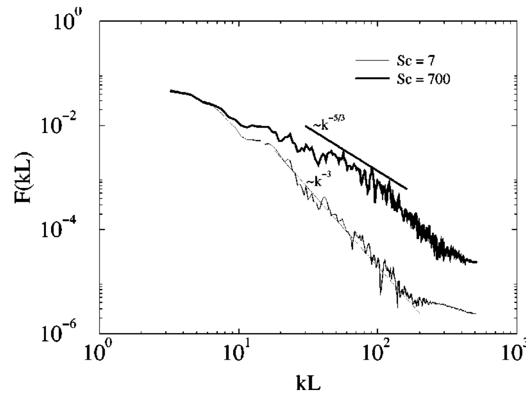

FIG. 7. Sc dependency on scalar spectra at low Re_L. The slopes for the straight lines shown are chosen from the experimental study by Huq and Britter (Ref. 44). LEM predictions of the spectral slopes are close to these experimental values.

convective range is seen. It is illustrated in Fig. 6(a) using a line with logarithmic slope of $-5/3$. Also shown is the presence of a viscous convective range with a k^{-1} form of the spectrum (as suggested by Batchelor[10]). For the viscous-diffusive range of scales, the LEM prediction does not agree with Batchelor's hypothesis. However, the exponential fall off of the scalar spectrum in this range, as proposed by Batchelor,[10] has never been established experimentally. In fact, theoretical analysis by Kraichnan[11] and DNS by Kerr[12] show that the spectrum drop off with increasing wave number is much slower than the exponential function hypothesized by Batchelor.

At low Sc, Batchelor *et al.*[9] proposed an inertial-convective range followed by a $k^{-17/3}$ form of inertial-diffusive range for the scalar spectrum. LEM, at Sc$=0.04$, produces a spectrum [Fig. 6(b)] that seems to scale as k^{-4} at high wave numbers. Earlier work by Kerstein[4] showed a $k^{-14/3}$ spectrum. There is no way to ascertain the validity of any of these scalings. To the extent of the current authors' knowledge, there seems to be no general agreement on the high wave number spectra at low Sc. Earlier, Kerstein[4] has argued that the instantaneous straining of the scalar field using discrete triplet maps creates a numerical artifact at low Sc. This argument is better illustrated using the observations made by Batchelor *et al.*[9] They argue that the local diffusion relaxation time (in spectral space) scales as $D^{-1}k^{-2}$ and hence, is much smaller than the local eddy turnover time $\epsilon^{-1/3}k^{-2/3}$ at large k (note that D is quite large for low Sc). In LEM, it is not possible to mimic this feature since the effect of an eddy on the scalar field is instantaneous.

C. Prediction of differential diffusion

Simultaneous prediction of the mixing of two different scalars in turbulence is important for mixing models from an application point of view. The most extensive application of mixing models is in reacting flow phenomena. In such flows, mixing of several species has to be done simultaneously. In other applications, where only one chemical species is of concern, one may still need to model heat diffusion in tandem. Huq and Britter[44] have conducted an experimental study on the turbulent diffusion of two passive scalars with Sc of 7 and 700. They have determined the Kolmogorov length scale in their experiment to be about 1 mm, and the square mesh (grid) they use to generate turbulence has openings that are 3.2 cm in each direction. To model this experiment, the integral length scale is set equal to 3.2 cm and η to 1 mm. The ratio of length scales is 32, which gives an integral scale Reynolds number of around 100. Given that the separation of length scales is not too high, one cannot expect a visible inertial range in energy spectra. In the experimental results, the one-dimensional velocity spectra shows less than a decade of wave number region with $k^{-5/3}$ scaling. The one-dimensional spectral density also shows a sharp decline close to k_η (the Kolmogorov wave number). Nevertheless, the linear eddy model is used to simulate the mixing of the two scalar fields at this low Reynolds number.

The linear eddy model predictions for the scalar spectra are shown in Fig. 7, along with lines indicating the spectral scaling found in experiments.[44] Also, these scalar spectra are plotted using the data from the same simulation (i.e., both scalar fields are stirred by the same turbulence field as in the experiments). Surprisingly, the model is able to accurately predict the scalar spectra even at such low Re.

The inertial convective scaling at Sc of 700, though surprising at this low Re, is more a consequence of independence of large scale mixing and small scale scalar dissipation. For this case, Peclet number (Pe) is much larger than the Reynolds numbers. The effect of viscosity on the inertial range energy cascade is much higher than the effect of molecular diffusivity on the inertial range scalar mixing. Therefore, it seems possible that a wider $k^{-5/3}$ range could be found in the scalar spectrum than in the velocity spectrum. Also, it is reasonable to expect that it is Pe and not Re that determines the presence of $k^{-5/3}$ scaling for scalars. As a result, this scaling is not found for the scalar field with lower Pe (i.e., lower Sc). The ratio of Kolmogorov length scale to Batchelor scale in Sc=700 case is only about 26 and hence, no viscous-convective scaling is seen in the spectrum though the Sc is reasonable high.

It is seen from Fig. 7 that with the stirring (effect of turbulent eddies) being the same, the scalars characterizing the fluid composition could have different spectra, hence different rates of turbulent diffusion. The only way differential diffusion of species occurs is due to molecular effects. There is evidence to suggest that molecular diffusion has nonvanishing effects on turbulent diffusion even at very high Re, where the stirring would be very high. The molecular effects are captured in LEM using a deterministic model and hence, their effect on differential diffusion should be accurately captured.

DNS of turbulence is now emerging as a very useful tool to study differential diffusion quantitatively. Although restricted to low Reynolds number flows in the past,[45,46] some of the more recent work[1–3] is at high enough Reynolds numbers to derive useful inferences about scalar diffusion. The specific issue of differential diffusion, in the context of DNS, was addressed by Yeung and Pope,[46] Yeung and Luo,[1] and Yeung.[3] Given the unphysical and widely different initial transients in LEM and DNS, the nonstationary simulations of Yeung and Pope[46] are less suitable for comparing the two methods. Yeung and Luo[1] and Yeung[3] include results for scalar fields with mean gradients in a statistical stationary fluid flow. The results from these two studies are used as references in the current study.

At $\mathrm{Re}_\lambda=230$, turbulent diffusion of two scalars with Sc of 1 and 0.125, respectively, is simulated using LEM. For quantifying differential diffusion between two scalars with concentrations ϕ_1 and ϕ_2 (with diffusivities D_1 and D_2, respectively), correlations between scalar concentrations ρ_{12} and their gradients g_{12} are computed:[1]

$$\rho_{12}=\frac{\langle \phi_1\phi_2\rangle}{[\langle \phi_1\rangle^2\langle \phi_2\rangle^2]^{1/2}}, \tag{13}$$

$$g_{12}=\left\langle \frac{\partial\phi_1}{\partial x}\,\frac{\partial\phi_2}{\partial x}\right\rangle\left[\left\langle\frac{\partial\phi_1}{\partial x}\right\rangle^2\left\langle\frac{\partial\phi_2}{\partial x}\right\rangle^2\right]^{-1/2}. \tag{14}$$

Using the observation that the mechanical to scalar time scale ratio τ_v/τ_s (discussed earlier) is fairly independent of molecular diffusivity, Yeung and Luo[1] arrive at an approximate expression for the gradient correlation, given as

$$g_{12}=2\left[\frac{D_1}{D_2}+\frac{D_2}{D_1}+2\right]^{-1/2}. \tag{15}$$

When the scalar field is locally isotropic, an expression for ρ_{12} is obtained in terms of g_{12} and the ratio of mechanical-to-scalar time scale ratios $(r_{1,2}=[\tau_v/\tau_s]_{1,2})$:

$$\frac{g_{12}}{\rho_{12}}\sim\frac{\sqrt{D_1D_2}(r_1+r_2)}{(D_1+D_2)\sqrt{r_1r_2}}. \tag{16}$$

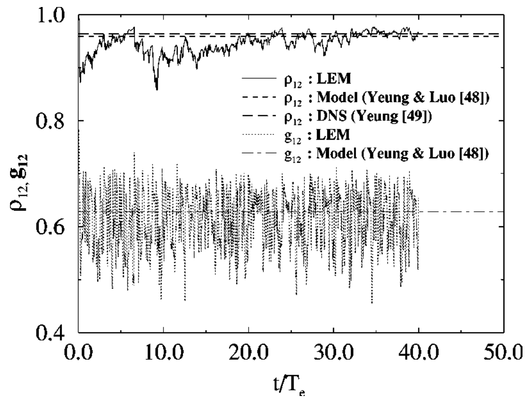

FIG. 8. Evolution of correlation coefficients for Sc=(1,1/8), $\mathrm{Re}_\lambda=230$.

Equation (15) seems to be valid at all Reynolds numbers (possibly beyond a certain lower limit). The DNS results at Re_λ of 90.0, 160[1] and 230[3] are found to establish this fact beyond a reasonable doubt. Model values for ρ_{12} [from Eq. (16)], on the other hand, seem to be valid at higher Re. At Re_λ of 230, the DNS and the model values of both g_{12} and ρ_{12} (from the two equations above) are within 0.5% of each other.

LEM predictions of g_{12} and ρ_{12} evolution at Re_λ of 230 are shown in Fig. 8. In this figure, T_e denotes the large eddy turnover time. Steady state values of ρ_{12} from Eq. (16) and DNS[3] are indicated as straight lines. The g_{12} computed using Eq. (15) is very close to the corresponding DNS value and is omitted. The correlation computed using the algebraic models of Yeung and Luo[1] and the linear eddy model are found to be in very good agreement with the corresponding DNS predictions.

The time variations of g_{12} and ρ_{12}, as shown by Yeung and Luo[1] and Yeung,[3] seem to be less intermittent than the variations in Fig. 8. While the difference in intermittency in DNS and LEM could be a contributing factor, one reason for this could also be the fact that, at a given instant, DNS provides a larger number of data points to compute these correlations than the linear eddy model. However, the (time) averaged values of the LEM predicted correlations match the corresponding values from DNS (and the models) with less than 4% difference.

The difference spectrum $F_z(k)$ (here, $z=\phi_1-\phi_2$), nondimensionalized using inertial convective scaling laws (using the χ of the less diffusive scalar), is shown in Fig. 9. While the location of the peak in the scaled spectrum is difficult to point out, an approximate location at $k\eta=0.344$ is shown in Fig. 9. This is close to the DNS prediction[3] of approximately 0.4. While the intermediate and high wave number spectrum seems to correlate well with the corresponding prediction by Yeung,[3] the low wave number spectra are quite different.

Yeung[3] has argued that the backscatter effects (in addition to intermittency) are quite different in DNS and LEM and hence, one may not find an agreement between the two approaches at low wave numbers. The backscatter effects are undoubtedly very different in both approaches. The backscatter is a phenomenon associated with nonlocal scale (triad) interactions. The scale interactions in LEM are weak,

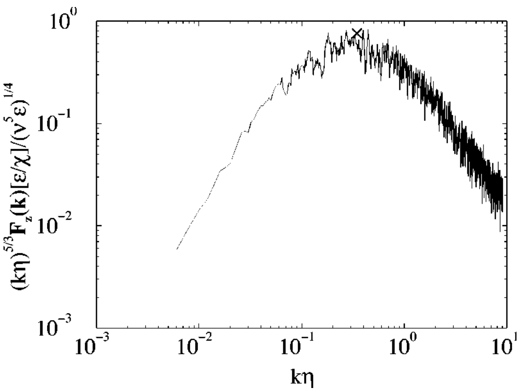

FIG. 9. Difference spectrum for Sc=(1,1/8) scaled using inertial-convective scaling of Sc=1 scalar.

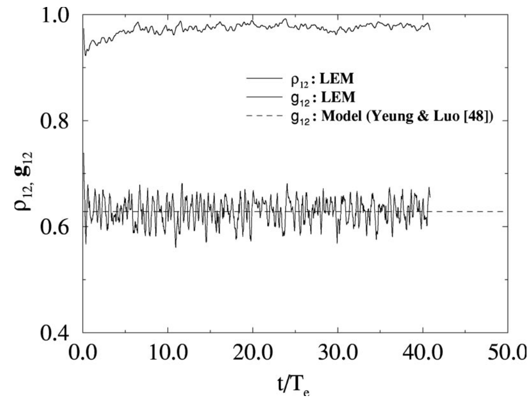

FIG. 11. Evolution of correlation coefficients for Sc=(1,8), $Re_\lambda=230$.

since each triplet map corresponds to the effect of a single eddy (of a given size) and is completely independent of other triplet maps (occurring either before or after the map in question). However, the fact remains that the wave number characteristics of both LEM and DNS are in question. The LEM considers the effect of stirring by scales only below the size of integral length scale. To model stirring by larger scales, the nature of the spectrum at low wave numbers needs to be considered in the design of LEM. As discussed earlier, in the context of predictions of scalar decay times, the forcing scheme in DNS could also have some impact on the results.

Since LEM is seen to work at Re much beyond the scope of direct simulations, the validity of the Eq. (15) for g_{12} at high Re can easily be assessed using LEM. This exercise would be of significant consequence in engineering model development since a binary fluid mixture with Sc ratio of 1/8 is representative of H_2–air mixtures often encountered in engineering. The scalar correlation and scalar gradient correlation at Re_L of 40 000 (about 11 times higher than in the case of Fig. 8) are shown in Fig. 10. The Yeung and Luo[1] model value for g_{12} is also shown as a straight line. The mean prediction of g_{12} is found to be 0.6335, which is different from the model value of 0.6285 by less than 0.8%.

The expression for g_{12} is a symmetric function of D_1/D_2 and D_2/D_1. This needs to be tested for the case when one of the scalars has Sc larger than 1.0. Under similar conditions as before, the mixing of two scalars with Sc of 1.0 and 8.0 is simulated. Small scale contribution to the gradient correlations (in turbulence) is quite significant. Since the small scale spectra is dependent on the value of Sc, this case is in no way similar to be previous case (i.e., the symmetry of the expression for g_{12} with respect to D_1/D_2 and D_2/D_1 is not obvious). The results are plotted in Fig. 11. The model value for g_{12} [from Eq. (15)] is also shown. It is seen, once again, that the LEM and Yeung's model predictions are in agreement.

To further test the Yeung and Luo model[1] for widely separated Sc, turbulent mixing of two scalars, with Schmidt numbers of 1.0 and 0.001, at Re_L of 40 000 ($Re_\lambda \sim 775$), is simulated. The predicted correlations are shown in Fig. 12, along with the value computed from Eq. (15). There is a reasonable match (to within 12%) even at an arbitrarily higher (than DNS) Re with wide Schmidt number difference. It was assumed that the mechanical-to-scalar time scale ratios are weak functions of diffusion in deriving the model.[1] In this case (as predicted by LEM), owing to the large difference in diffusivities, the ratios turn out to be different for the two species by about 22%. However, fluids with Sc much less than 0.1 are seldom encountered in nature and hence, the algebraic model for g_{12} provided by Yeung and Luo[1] can be considered a good approximation for most situations.

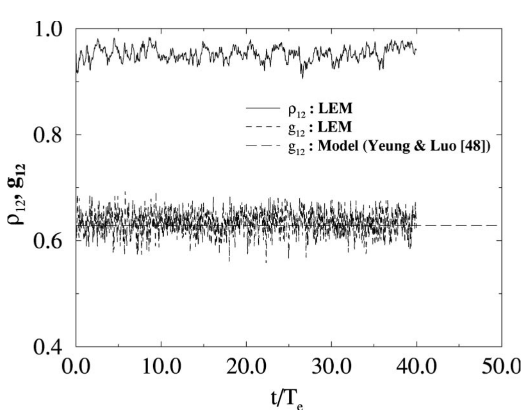

FIG. 10. Evolution of correlation coefficients for Sc=(1,1/8), $Re_\lambda=775$.

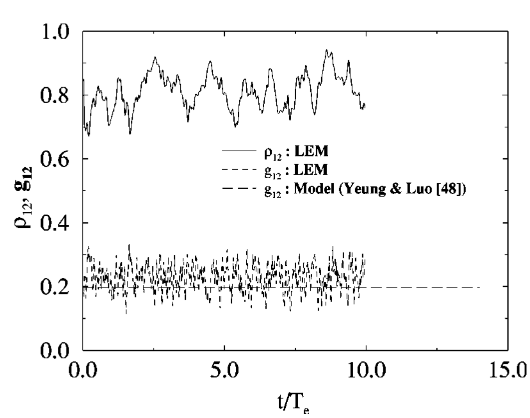

FIG. 12. Evolution of correlation coefficients for Sc=(1,0.001), $Re_\lambda=775$.

IV. CONCLUSIONS

The linear eddy model is used here to predict scalar mixing over a wide range of parameter space. No empirical inputs are needed to match predictions with experiments or DNS studies. The results of this LEM study point to the following observations.

Scaling laws provide only qualitative information about the relative distribution of mechanical or scalar energy among various wave numbers. The intensity of the spectra (which determine the total energy and the total scalar variance) depend on the numerical constants such as α and β. It is not clear if the scatter in the values of α and β are a consequence of deviations from the expected $k^{-5/3}$ scaling. If α and β indeed depend on the deviations from expected scaling, then their values are most likely not correlated. This is due to the fact that the intermittency patterns (which in part, determine the scaling) in energy spectrum and the scalar spectrum are often quite different and are independent of each other. On the other hand, from a purely deterministic point of view, the amplitude levels of scalar fluctuations (quantified by β) depend on the extent of stirring by the inertial range fluid dynamics which depends on α. It is, therefore, probable that the scatter in the values for the Obukhov–Corrsin constant could be related to the scatter in values of the Kolmogorov constant, i.e., higher and lower values of β, respectively, correlate with higher and lower values of α.

For the modeling approaches such as LEM, the Obukhov–Corrsin constant indeed has a monotonic relationship with the Kolmogorov constant. Whether the experimental data supports this argument remains an open question. The LEM, however, predicts a value of 0.7 for the Obukhov–Corrsin constant for a Kolmogorov constant of 1.5. These numbers were generally seen as the median values of the two constants in independent studies.[34,37]

It is seen that ratio of time scales characterizing the velocity decay and the scalar decay is either a nonmonotonic function of Re or depends on another parameter. This has been established in experiments.[39] Although no explanation exists for the trends in time scale ratio, its variation seems to be captured by LEM.

The form of the scalar spectrum at the viscous-diffusive scales in a high Sc fluid flow still needs to be established experimentally. The LEM predictions seem to agree with Batchelor's viscous-convective scaling. In the viscous-diffusive range, LEM results are in qualitative agreement with previous theoretical[11] and DNS[12] results (i.e., slower fall in the spectrum than the exponential predictions of Batchelor). In case of low Sc also, the LEM predicts correctly an inertial-convective range. Differential diffusion effects are also shown to be correctly captured by LEM and the simulation results confirm the viability of the models proposed by Yeung even in high Re flows beyond the range of DNS.

Finally, it can be concluded that the LEM provides an efficient and accurate tool for analyzing scalar mixing in turbulent fluids. By using a reaction-diffusion equation instead of the diffusion equation, this approach can also be used to model turbulent combustion.[21,22] In premixed combustion, Smith and Menon[21] used LEM to predict the dependence of turbulent flame speeds on the turbulent intensities in the flamelet regime. This work has been extended to the distributed (thin-reaction-zones) regime using a detailed chemical kinetics model.[47] The present effort is more applicable to nonpremixed combustion and study is underway to investigate the effect of scalar mixing in the presence of chemical reactions and heat release.

ACKNOWLEDGMENTS

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